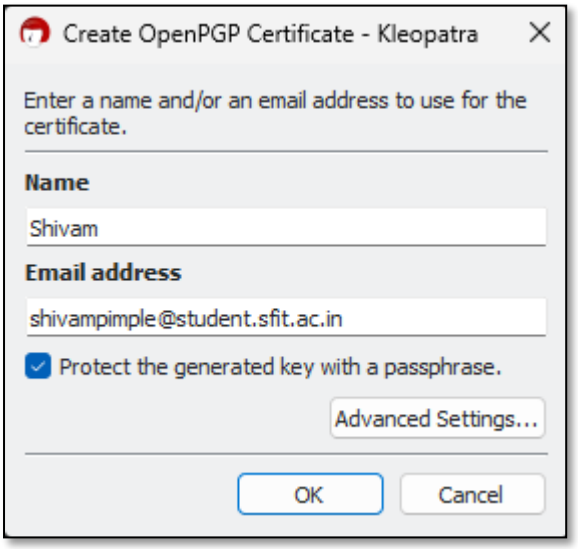
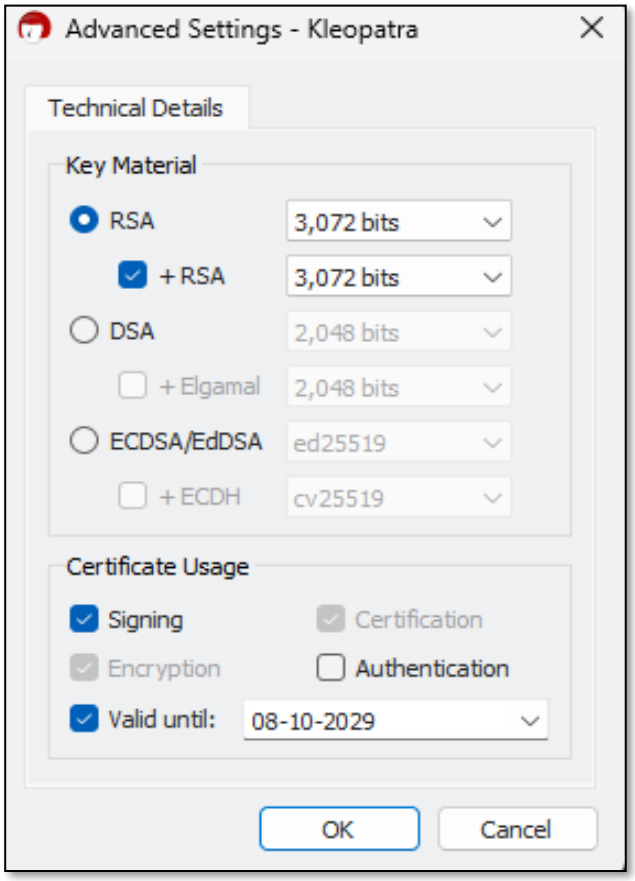
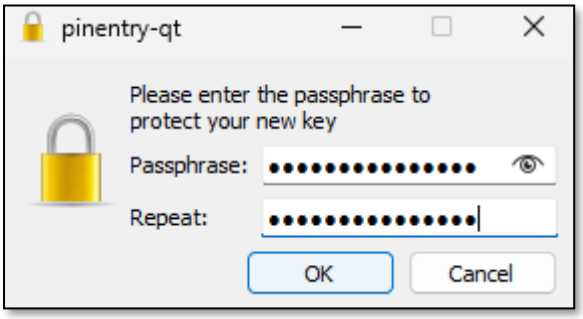
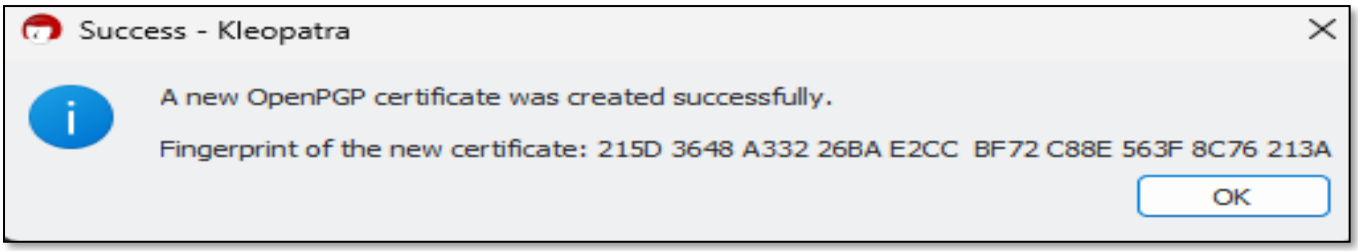
Experiment 11:

Step 1: Create a new key pair







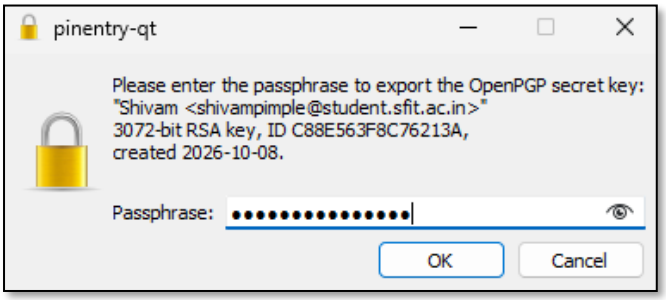


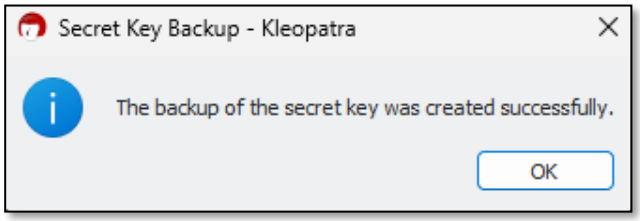
Anushka	**anushkachaugule95@student.sfit.ac.in**	**certified**	**06-10-2026**	**06-10-2029**	**57E4 7AED 89BF C000**
Jaans Pravin	factomedia15@student.sfit.ac.in	certified	06-10-2026	06-10-2029	F865 DCB4 DCC2 932E
PranitRane	**pranitrane.2005@student.sfit.ac.in**	**certified**	**10-10-2025**	**10-10-2028**	**CA2A A11A A857 23F7**
Shivam	**shivampimple@student.sfit.ac.in**	**not certified**	**08-10-2026**	**08-10-2029**	**C88E 563F 8C76 213A**
SohamPatil	**soham.pat153@student.sfit.ac.in**	**certified**	**09-10-2025**	**09-10-2028**	**7876 5955 64DC DD88**
Student	**makandarsaud@gmail.com**	**certified**	**05-10-2026**	**05-10-2029**	**9EE3 944E 7DBC 5F3F**
Sumit	Sumitbhendigeri1985@student.sfit.ac.in	certified	05-10-2026	05-10-2029	0EC5 3D31 6B04 5E93
Varun	pednekarvarun01@student.sfit.ac.in	certified	10-10-2025	10-10-2028	3BD9 1218 E56C 0F7E

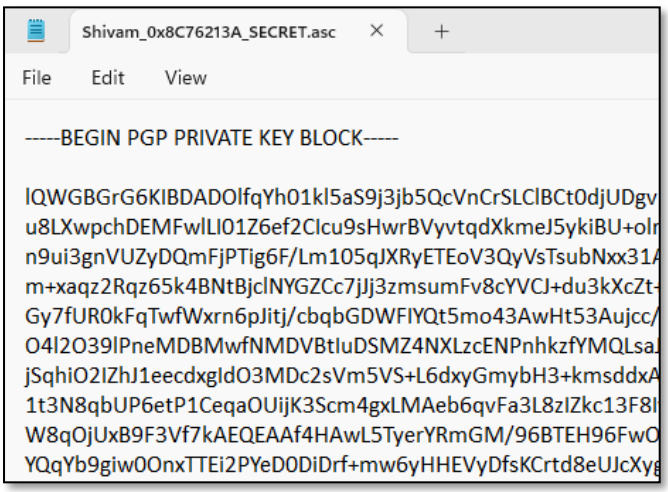
Observation:

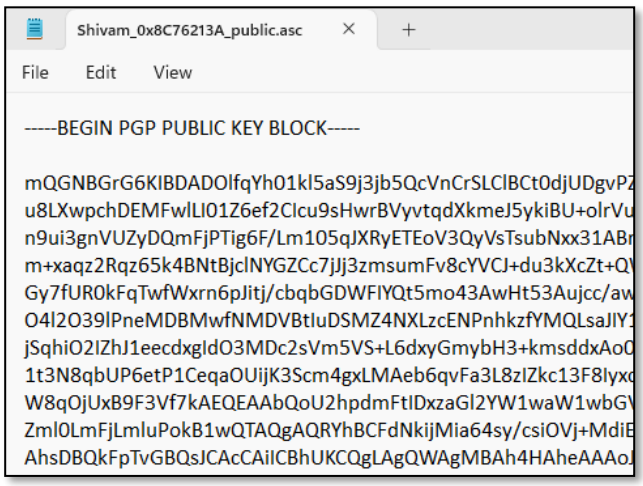
- The name and email address were entered in Kleopatra to create a new OpenPGP certificate.
- RSA key settings, certificate usage, and validity period were configured.
- A passphrase was entered and confirmed to protect the secret key.
- Kleopatra displayed a message confirming successful certificate creation.
- The newly created certificate appeared in the Kleopatra certificate list.

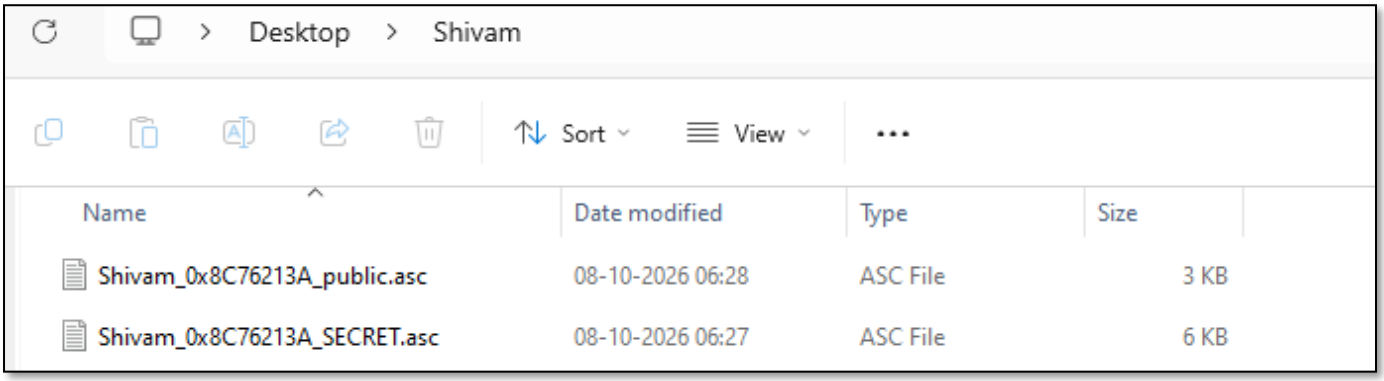
Step 2: Save the keys on the desktop and observe the keys.









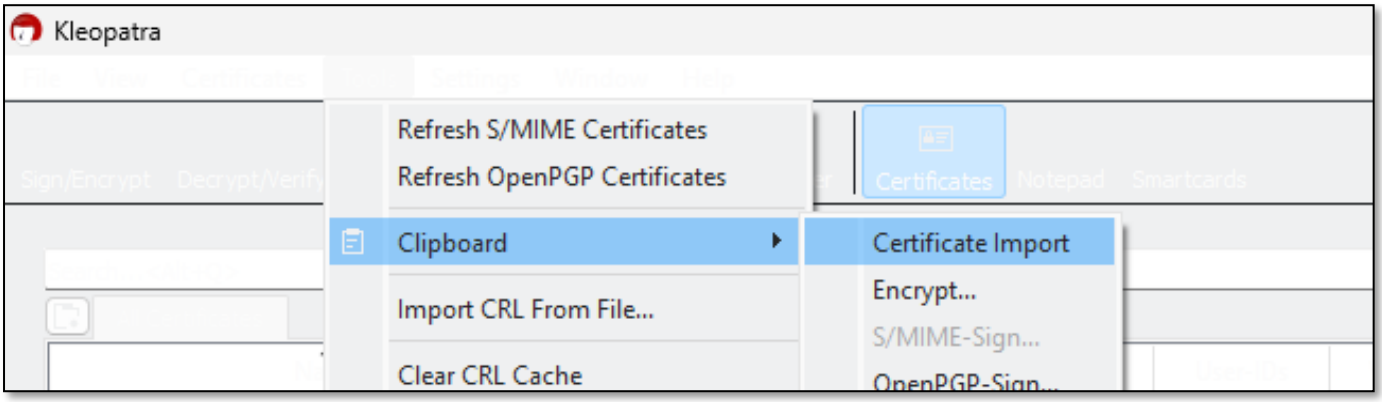


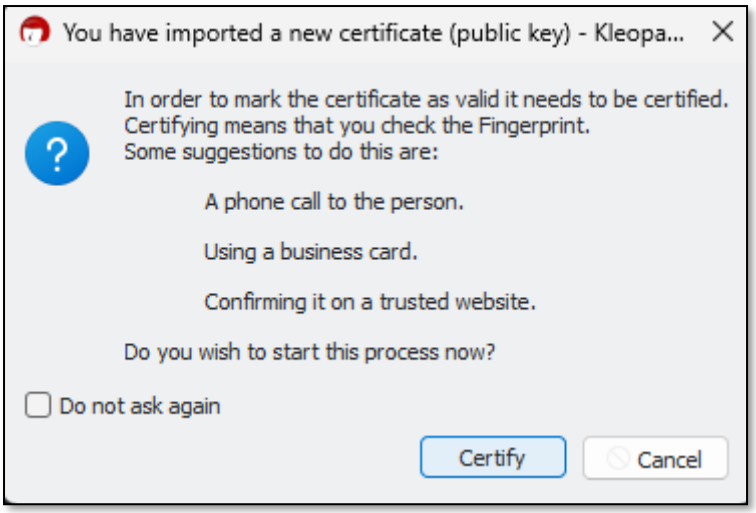
Observation:

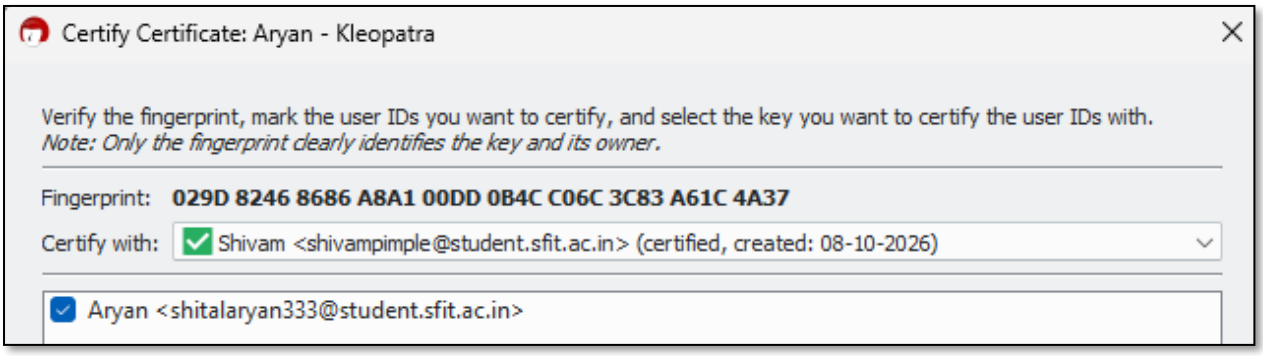
- The passphrase was entered to authorize the secret key export.
- Kleopatra confirmed that the secret key backup was created successfully.

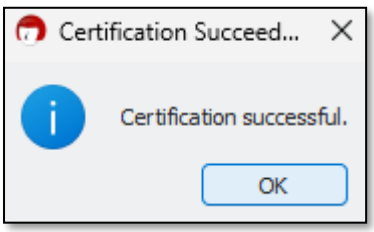
- The exported secret key was displayed as an .asc file.
- The exported public key was displayed as an .asc file.
- Both key files were successfully saved on the desktop.

Step 3: Import your secret communication partner's (your friend's) key in Kleopatra.









Observation:

- The Certificate Import option was selected in Kleopatra.
- The imported certificate was detected and the fingerprint verification and certification process was initiated.
- The communication partner's certificate was successfully certified.

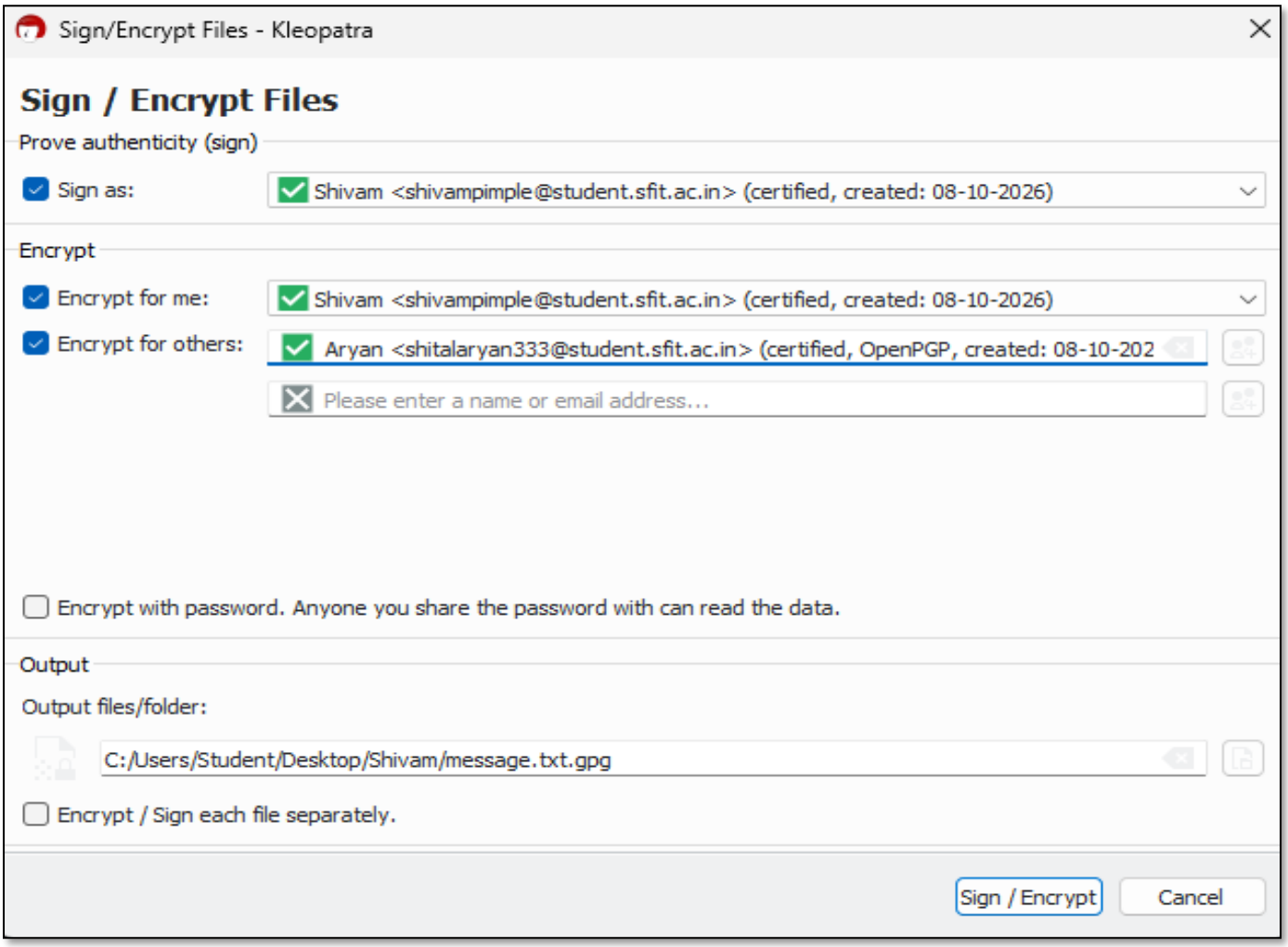
Step 4: Write your secret message in any word processing software.

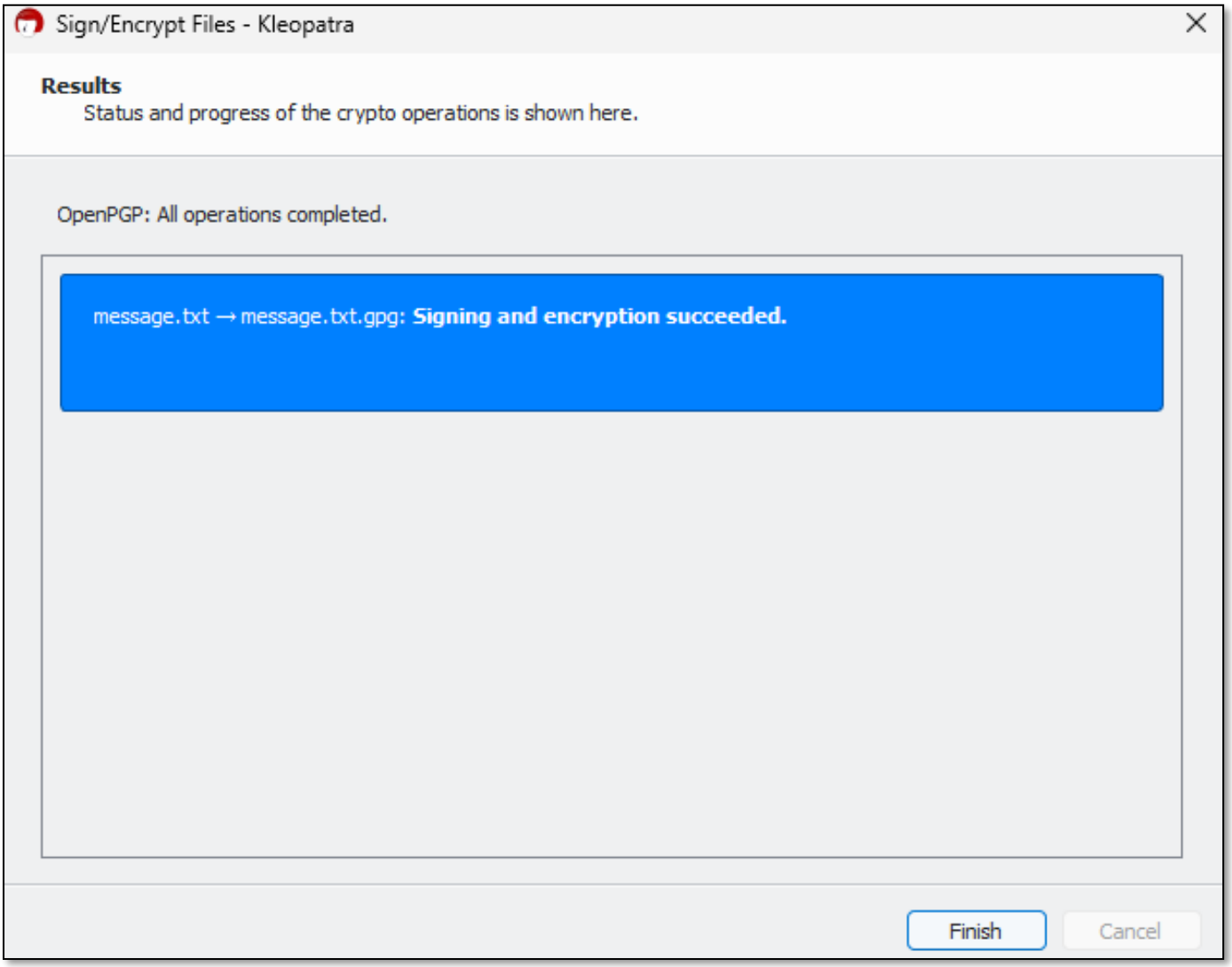


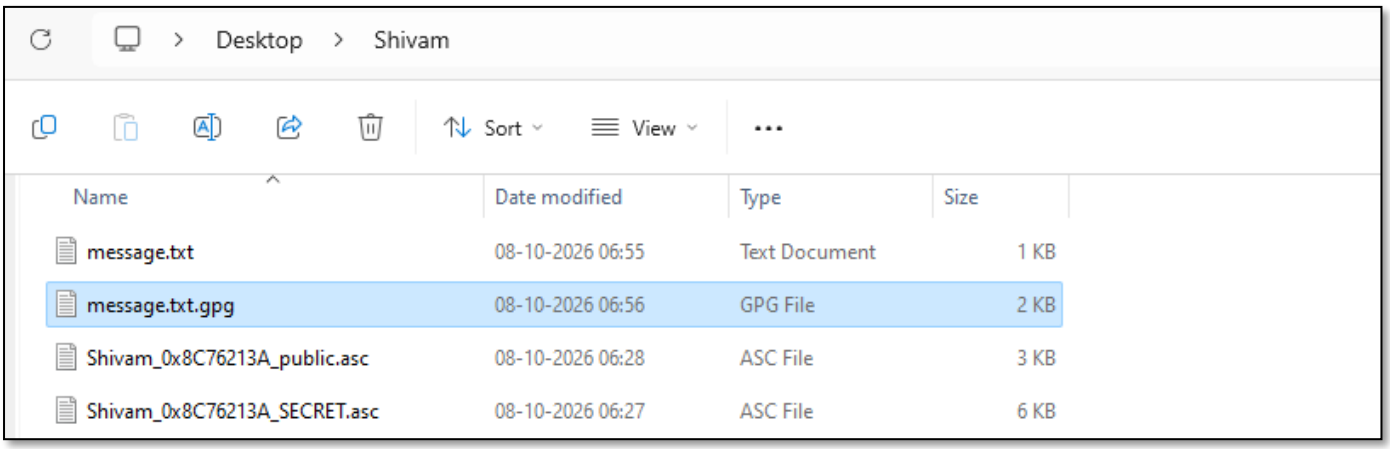
Observation:

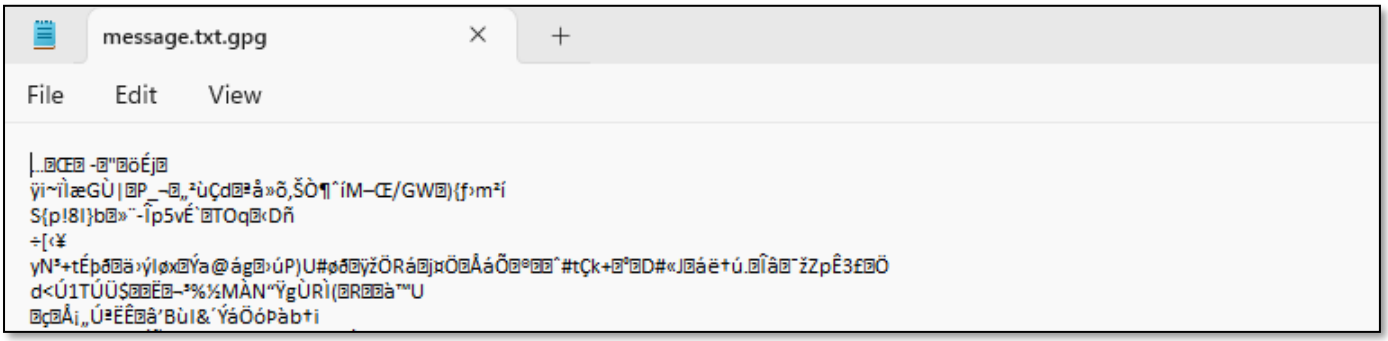
- A secret message was written in a text editor and saved as a text file for encryption.

Step 5: Encrypt this message file with recipient's public key. Save and send it to the recipient through any preferred communication medium.




Sign/Encrypt Files - Kleopatra
Results
Status and progress of the crypto operations is shown here.
OpenPGP: All operations completed.
message.txt → message.txt.gpg: Signing and encryption succeeded.
Finish
Cancel

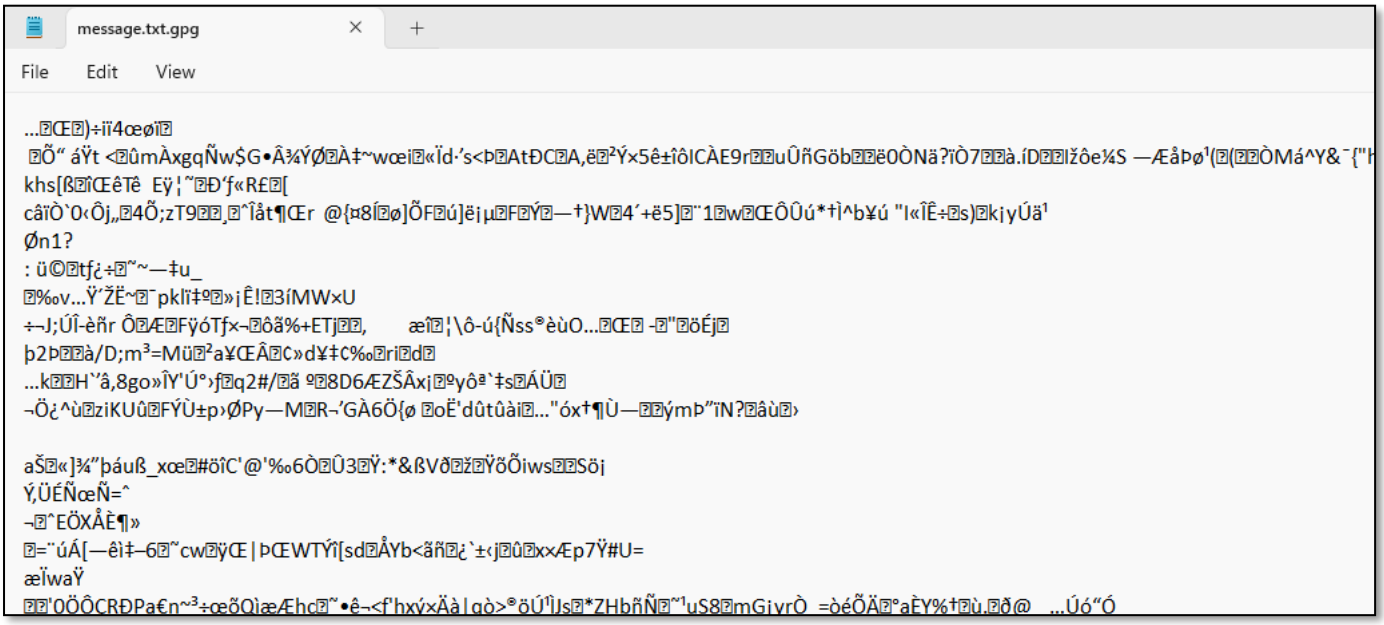

Desktop
Shivam
Sort
View
Name
Date modified
Type
Size
message.txt 08-10-2026 06:55 Text Document 1 KB
message.txt.gpg 08-10-2026 06:56 GPG File 2 KB
Shivam_0x8C76213A_public.asc 08-10-2026 06:28 ASC File 3 KB
Shivam_0x8C76213A_SECRET.asc 08-10-2026 06:27 ASC File 6 KB

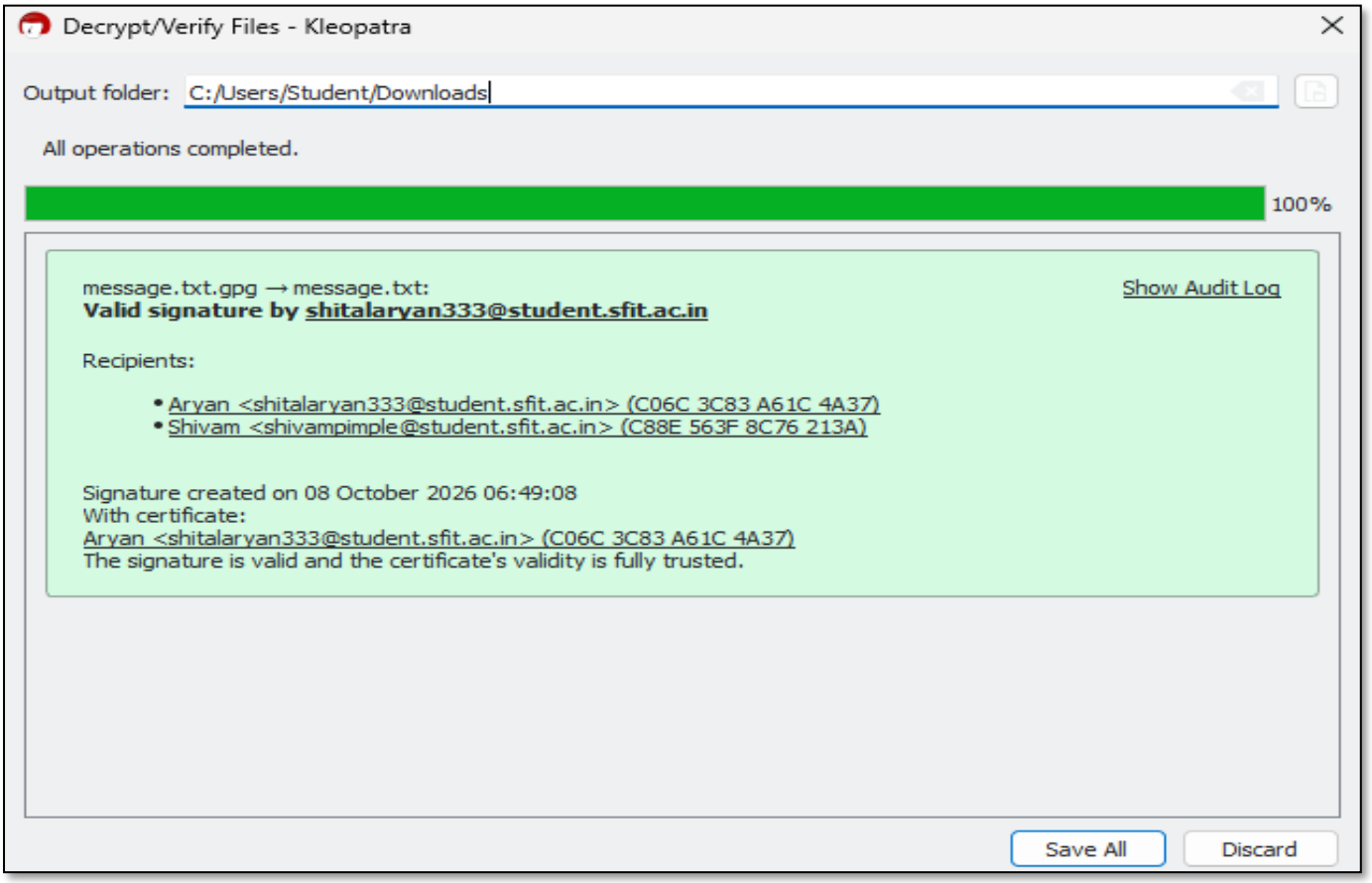

message.txt.gpg
File
Edit
View

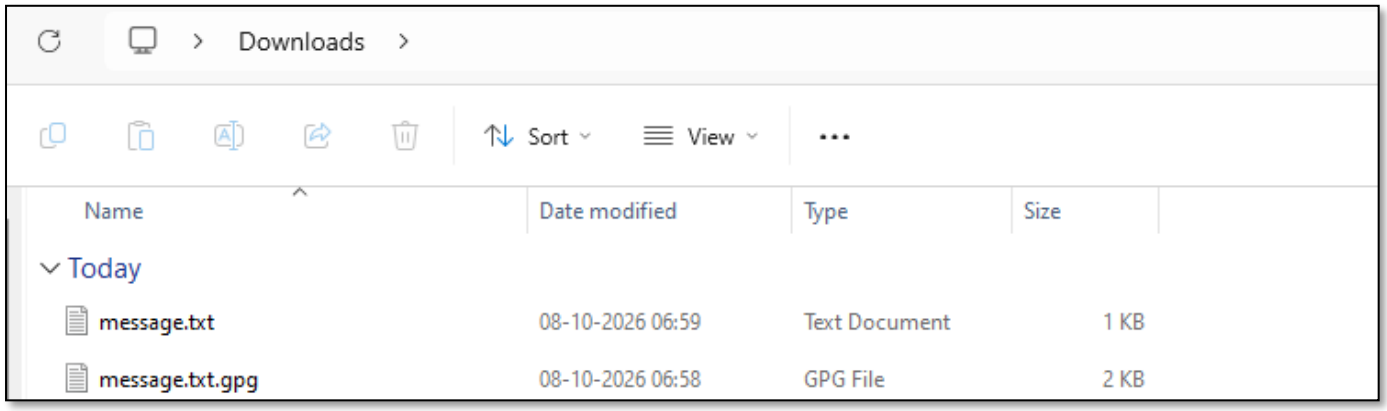
Observation:

- The sender's key and recipient's public key were selected in the Sign/Encrypt Files window.
- Kleopatra confirmed that signing and encryption were completed successfully.
- The encrypted .gpg file was generated and saved along with the original message and key files.

Step 6: Decrypt and observe the message received by you.









Observation:

- The encrypted .gpg file was opened and displayed as unreadable encrypted data.
- Kleopatra successfully decrypted the file and verified the digital signature.
- The decrypted message.txt file was generated successfully.
- The original readable message was recovered after decryption.